

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: (II)

Problem Set 2

(Exponential family of distributions, Completeness, UMVUE)

1. For each of the following, verify whether they are members of the exponential family. If they are, identify a sufficient statistic, obtain the minimal canonical form, and verify whether they are full rank exponential families. (Assume all parameters are unknown and in their usual space, and distributions are defined in their usual supports, unless specified otherwise.)

- (a) Geometric distribution: $p_\theta(x) = \theta^{x-1}(1 - \theta)$
- (b) Poisson distribution: $p_\lambda(x) = e^{-\lambda} \frac{\lambda^x}{x!}$
- (c) t-distribution with v degrees of freedom
- (d) Beta distribution: $B(\alpha, \beta)$
- (e) Binomial distribution: $Bin(n, \theta)$
- (f) Normal distribution: $N(\mu, \sigma^2)$
- (g) Normal distribution with constrained parameters: $N(\theta, \theta^2)$, where $\theta > 0$.
- (h) Gamma distribution: $f_{\alpha, \beta}(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, where $\alpha, \beta > 0$ and $x \geq 0$.
- (i) The distribution of the random variable $Y = \theta \ln X$, where $\theta > 0$ is known and X has the p.d.f.:

$$f_X(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} e^{-x/\beta} x^{\alpha-1}, \quad x > 0$$

where α is known.

- (j) Power series distribution with p.m.f.:

$$p_\theta(x) = P(X = x) = \frac{\gamma(x)\theta^x}{c(\theta)}, \quad x = 0, 1, 2, \dots$$

where $\gamma(x) \geq 0$, $\theta > 0$, and $c(\theta) = \sum_{x=0}^{\infty} \gamma(x)\theta^x$.

2. Let $X_1, \dots, X_n$ be i.i.d. random variables with p.d.f.

$$\exp \left\{ - \left(\frac{x - \mu}{\sigma} \right)^4 - \vartheta(\theta) \right\}$$

where $\theta = (\mu, \sigma^2) \in \Theta = \Re \times \Re^+$.

- (a) Show that the family $\{p_\theta : \theta \in \Theta\}$ is an exponential family, where p_θ is the joint distribution of $\{X_1, \dots, X_n\}$.
 - (b) Obtain the minimal canonical form and the natural parameter space corresponding to the minimal canonical form.
3. Suppose we draw n items using simple random sampling without replacement from a population of N items containing D defective items. Define

$$X_i = \begin{cases} 1, & \text{if the } i\text{-th drawn item is defective} \\ 0, & \text{otherwise} \end{cases}$$

Prove or disprove that $T = \sum_{i=1}^n X_i$ is complete.

4. Let $X_1, \dots, X_n$ be a random sample from each of the following family of distributions. Verify whether the minimal sufficient statistic is complete or not:

- (a) $U(\theta - 1/2, \theta + 1/2)$, $\theta \in \Re$
 - (b) $Exp(\theta, 1)$, $\theta \in \Re$
 - (c) $Cauchy(\theta, 1)$, $\theta \in \Re$
5. $(X_1, Y_1), \dots, (X_n, Y_n)$ are independent and identically distributed from a bivariate normal distribution with $E[X_i] = E[Y_i] = 0$, $Var(X_i) = Var(Y_i) = 1$ and $Cov(X_i, Y_i) = \theta$. Prove or disprove that the MSS is complete.
6. Let $X_1, \dots, X_n$ be a random sample from $Bin(1, \theta)$, $\theta \in (0, 1)$. Find the UMVUE for each of the following parametric functions:
- (a) $g(\theta) = \theta$
 - (b) $g(\theta) = \theta^4$
 - (c) $g(\theta) = \theta(1 - \theta)^2$
7. $X_1, \dots, X_n$ is a random sample from $U(0, \theta)$, $\theta > 0$. Find the UMVUE of $g(\theta) = \theta^2$.
8. $X_1, \dots, X_n$ is a random sample from $N(0, \theta)$, $\theta > 0$. Find the UMVUE of $g(\theta) = \theta^2$.
9. Give an example for each of the following cases:
- (a) A statistic that is sufficient but not complete
 - (b) A statistic that is complete but not sufficient